WHO AUDITS THE JUDGE?

From Broken Instrument to Redesigned Protocol

Auditing an LLM-as-a-Judge Pipeline for RAG-Based Academic Mentorship

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CASE STUDY

An AI mentor for **early-career women faculty in Chilean higher education**, grounded in **72 interviews** and **53 papers**.

4 retrieval setups × 49 queries → **196 outputs** → **Two LLM judges**
Zero-retrieval · Standard · HyDE · Hierarchical | DeepSeek-V4-Flash | GPT-5.6 Luna + MiniMax M3

91.67%

faithfulness score the AI judge gave to a model that saw ZERO documents.

11 of 12 claims judged “supported” in one inspected zero-retrieval response: fluent memory mistaken for evidence.

An LLM judge can be consistent and still be wrong. Audit the judge before you trust the benchmark.

V1.0 · What our audit found

Five measured pathologies (plus 7 structural rubric flaws)

86.1%	Author names from a background note became a hidden grading checklist (GPT 334/388; MiniMax 58.8%)
96 / 97	Zero-retrieval rationales capped at Level 3 for not citing those two authors
80–98%	of scores piled up at Level 4: ceiling collapse erased differences
r = .35–.39	Length bias, despite explicit “ignore length” instructions
ICC = .19	Negligible convergent validity with RAGAS metrics

REDESIGN →

V1.1 · Four architectural firewalls

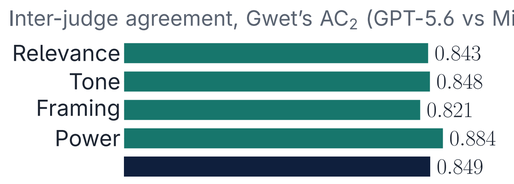
Structure instead of prompt instructions (Design Science Research)

- **Layer 0** — **Triage + N/A masks** set before scoring: Framing N/A for 32.7%, Power N/A for 61.2% of queries
- **Layer 1** — **Atomic claim verification** checks grounding claim by claim, isolated from quality
- **Layer 2** — **Decontaminated BARS rubric**: 4 dimensions, behavioral anchors, scale-inversion control
- **Layer 3** — **Care routing**: information gaps flagged to Help Desk or Ethics Committee
- **Report** — **Disaggregated reporting**: weighted Gwet’s AC_2 + bootstrap 95% CI; length as covariate

✓ Human calibration (N = 40): Gwet’s AC_2 = 0.73–0.82 across all 4 dimensions

Judge consensus can be a mirage.

Inter-judge agreement, Gwet’s AC_2 (GPT-5.6 vs MiniMax)



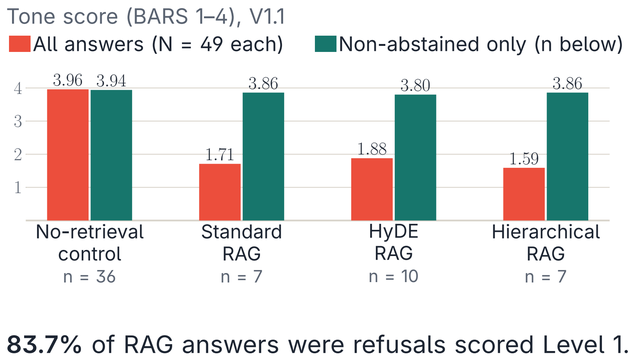
Rationales enforcing the leaked author checklist



Two model families agreed strongly because both obeyed the *same leaked instruction*. **Agreement is not validity.**

RAG looked 2 points worse. It was abstaining.

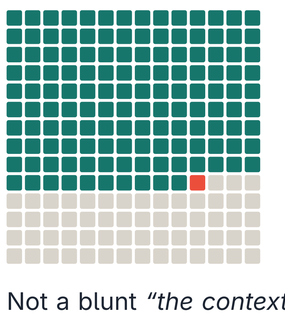
Tone score (BARS 1–4), V1.1



83.7% of RAG answers were refusals scored Level 1. Controlling for length, RAG’s tone effect turns slightly positive (OLS, R^2 = 0.99). Relevance still lags; small n, indicative.

Missing context is a relational signal.

Each square = 1 of 196 outputs (Layer 3, V1.1)



137

of **196** outputs flagged and routed to human support (**69.9%**)

- **136** Help Desk / Ombudsperson
- **1** Academic Ethics Committee
- **59** answered directly

Not a blunt *“the context does not contain the answer”*: we flag the gap and point to a human channel. Flags are logged; user-facing delivery not yet tested.

Audit your judge on a budget

Six cost-ranked controls for lean teams, about 8 hours in total

0	**Archive outputs + Git tags**	Enables paired before/after audits	free
1	**Zero-context negative control**	Hallucinated grounding (91.67%)	< $0.20
2	**Ex-ante applicability masks**	Forced-scale distortion	~1 h
3	**Non-abstained decomposition**	Refusals confounded with quality	~1 h
4	**Regex audit of rationales**	Prompt-metadata leakage (86.1%)	~2 h
5	**Expert human calibration (N ≈ 30)**	Judge leniency + misalignment	~4 h

Full paper + BARS rubric

Scan for 3-min challenge + PDF

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Detailed results and a live walk-through on the screen

Limits: one likely-unblinded annotator · Chilean Spanish corpus · V1.1 tuned on the same 196 outputs (out-of-sample pending)

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